TWO LIFTS INTO A BOUNDED BELOW COMPLEX OF INJECTIVES ARE HOMOTOPIC

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6713-6755, following the statement at 6697-6711.

Editorial note. No omissions: the direct argument is displayed in full, including the homotopy equations and the extension of the homotopy by zero on the second summand.

Allowed standard prerequisites (editorial).

- Reduction to termwise split monomorphisms. lemma-make-injective (same chapter, cited around line 6712). Provides tilde-alpha, pi and s with pi composed s the identity, letting one replace alpha by a termwise split inclusion.
- Acyclic complexes map trivially into injective complexes. lemma-acyclic-is-zero (same chapter, cited at the final step ~6755). Kills the difference morphism once it factors through the acyclic quotient M.
- Alternative route. remark-easier-proofs (same chapter, cited at the start ~6713). The author notes the result also follows from the remark; the direct argument is given in full anyway.

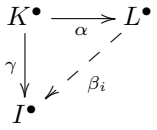
Source reference. Original lemma 013R: sources/stacks-derived/pinned-original-source.tex, source lines 6525–6525. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 013N: sources/stacks-derived/pinned-original-source.tex, source lines 2840–2840. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 05TF: sources/stacks-derived/pinned-original-source.tex, source lines 6568–6568. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Consider a solid diagram*



where $I^\bullet$ is bounded below and consists of injective objects, and α is a quasi-isomorphism. Any two morphisms β_1, β_2 making the diagram commute up to homotopy are homotopic.

Proof. This follows from Remark 05TF. We also give a direct argument here.

Let $\tilde{\alpha} : K \to \tilde{L}^\bullet$, π, s be as in Lemma 013N. If we can show that $\beta_1 \circ \pi$ is homotopic to $\beta_2 \circ \pi$, then we deduce that $\beta_1 \sim \beta_2$ because $\pi \circ s$ is the identity. Hence we may assume $\alpha^n : K^n \to L^n$ is the inclusion of a direct summand for all n. Thus we get a short exact sequence of complexes

$$0 \to K^\bullet \to L^\bullet \to M^\bullet \to 0$$

which is termwise split and such that $M^\bullet$ is acyclic. We choose splittings $L^n = K^n \oplus M^n$, so we have $\beta_i^n : K^n \oplus M^n \to I^n$ and $\gamma^n : K^n \to I^n$. In this case the condition on β_i is that there are morphisms $h_i^n : K^n \to I^{n-1}$ such that

$$\gamma^n - \beta_i^n|_{K^n} = d \circ h_i^n + h_i^{n+1} \circ d$$

Thus we see that

$$\beta_1^n|_{K^n} - \beta_2^n|_{K^n} = d \circ (h_1^n - h_2^n) + (h_1^{n+1} - h_2^{n+1}) \circ d$$

Consider the map $h^n : K^n \oplus M^n \to I^{n-1}$ which equals $h_1^n - h_2^n$ on the first summand and zero on the second. Then we see that

$$\beta_1^n - \beta_2^n - (d \circ h^n + h^{n+1}) \circ d$$

is a morphism of complexes $L^\bullet \to I^\bullet$ which is identically zero on the subcomplex $K^\bullet$. Hence it factors as $L^\bullet \to M^\bullet \to I^\bullet$. Thus the result of the lemma follows from Lemma 013R. □